

Part A, Topic 2 - Olympic Tweets



MONASH University

Lance Belen

May 23, 2025

Contents

1	Data Wrangling, Visualisation, and Analysis	3
1.1	Visualisation	7

1 Data Wrangling, Visualisation, and Analysis

First, load the packages needed.

```
options(repos = c(CRAN = "https://cloud.r-project.org/"))

if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse")
}
if (!requireNamespace("lubridate", quietly = TRUE)) {
  install.packages("lubridate")
}
if (!requireNamespace("impute", quietly = TRUE)) {
  install.packages("impute")
}
if (!requireNamespace("dplyr", quietly = TRUE)) {
  install.packages("dplyr")
}
if (!requireNamespace("magrittr", quietly = TRUE)) {
  install.packages("magrittr")
}
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2")
}
if (!requireNamespace("readxl", quietly = TRUE)) {
  install.packages("readxl")
}
if (!requireNamespace("stringr", quietly = TRUE)) {
  install.packages("stringr")
}
if (!requireNamespace("forcats", quietly = TRUE)) {
  install.packages("forcats")
}
if (!requireNamespace("tidytext", quietly = TRUE)) {
  install.packages("tidytext")
}

library(tidyverse)
library(lubridate)
library(impute)
library(dplyr)
library(magrittr)
library(ggplot2)
library(readxl)
library(stringr)
library(forcats)
library(tidytext)
```

Then, read and open the dataset.

```
tweets <- read.csv("Olympics_tweets.csv")
```

Next, explore the dataset.

```

cat("Number of rows: ", nrow(tweets), "\n")
cat("Number of columns: ", ncol(tweets), "\n")
knitr::kable(head(tweets))
cat("Structure of the dataset: \n")
str(tweets)
cat("Summary of the dataset: \n")
summary(tweets)

```

```
## Number of rows: 114213
```

```
## Number of columns: 13
```

id	text	user_name	screen_name	location	verified	description	user_created_at	tweet_created_at	language
1.418962418	Mirabai Chanu's maiden Olympic silver helped India clinch joint-12th spot on the medals tally on Day 1 atâ€¦ https://t.co/ksy8OvaJrl	ITGD_Nipitas	ITGD_Nipitas	India	FALSE	Live cricket scores- news-analysis and fun facts on all your favourite sports.	26/09/2020 10:57	24/07/2021 15:52	en
1.418962418	Mirabai Chanu brings home first silver medal on Day 1. DSEU congratulates her on this achievement. A brilliant stâ€¦ https://t.co/allIvBeujfH	dseu_official	dseu_official		FALSE	The official Twitter account for Delhi Skill and Entrepreneurship Univeristy.	12/11/2020 9:01	24/07/2021 15:52	en
1.418962418	Mirabai Chanu congratulations to Mirabai Chanu for starting the medal tally for India in the Tokyo Olympics 2020 by wiâ€¦ https://t.co/NukYganXeA	Sutej_Numar	Sutej_Numar	Delhi-India	FALSE	Believe in yourself @Yt3	20/04/2020 16:56	24/07/2021 15:52	en
1.418962418	Mirabai Chanu Congratulations to ace Indian weightlifter @mirabai_chanu for winning a Silver in the womenâ€™s #Weightliftingâ€¦ https://t.co/QJqHGstelZ	NsChandrapada	NsChandrapada	Telangana	FALSE	Govt Teacher YouTuber Ambedkarist Indian Cricket Lover Founder of @BORN4WIN Social Activist Kargil Vijay Diwas	8/08/2017 1:26	24/07/2021 15:53	en

id	text	user_screen_name	user_location	retweet_count	favorited	favorite_count	user_description	user_created_at	user_followers	user_friends	date	language
1.418963e+18	Congratulations to @mirabai_chanu for winning silver medal in weight lifting. This is first medal for India in Tokyo! https://t.co/NhVphf01A	Prabhat Kumar	Uttar Pradesh	0	FALSE	0	Nationalist Engineer Plastics Technologist 'CIPET'ian	14/11/2016	17:13	15:53	24/07/2021	en
1.418963e+18	congratulates India's Pride Mirabai Chanu on her winning ist T! https://t.co/4BizUPL60n	JaiHar	India	2560563	FALSE	0	MBA(MDU)- FORMER GM HARYANA	29/07/2018	16:19	15:53	24/07/2021	en

Structure of the dataset:

```
## 'data.frame': 114213 obs. of 13 variables:
## $ id : num 1.42e+18 1.42e+18 1.42e+18 1.42e+18 1.42e+18 ...
## $ text : chr "Mirabai Chanu's maiden Olympic silver helped India clinch joint-12th spot
## $ user_screen_name: chr "ITGDsports" "dseu_official" "SutejKumar4" "NsChandapaka" ...
## $ user_location : chr "Noida- India" NA "New Delhi- India" "Nalgonda- Telangana" ...
## $ retweet_count : int 1 0 0 1 0 0 0 1 0 0 ...
## $ favorited : logi FALSE FALSE FALSE FALSE FALSE FALSE ...
## $ favorite_count : int 1 1 0 0 1 0 0 2 0 0 ...
## $ user_description: chr "Live cricket scores- news- analysis and fun facts on all your favourite s
## $ user_created_at : chr "26/09/2017 10:57" "12/11/2020 9:01" "20/04/2020 16:56" "8/08/2017 1:26" .
## $ user_followers : int 9333 1168 72 5907 2721 1 61 7247 180 771 ...
## $ user_friends : int 28 4 143 60 353 84 405 185 9 1071 ...
## $ date : chr "24/07/2021 15:52" "24/07/2021 15:52" "24/07/2021 15:52" "24/07/2021 15:53
## $ language : chr "en" "en" "en" "en" ...
```

Summary of the dataset:

```
## id text user_screen_name user_location
## Min. :1.419e+18 Length:114213 Length:114213 Length:114213
## 1st Qu.:1.420e+18 Class :character Class :character Class :character
## Median:1.420e+18 Mode :character Mode :character Mode :character
## Mean :1.420e+18
## 3rd Qu.:1.421e+18
## Max. :1.422e+18
##
## retweet_count favorited favorite_count user_description
## Min. : 0.0000 Mode :logical Min. : 0.00 Length:114213
## 1st Qu.: 0.0000 FALSE:114210 1st Qu.: 0.00 Class :character
## Median : 0.0000 NA's :3 Median : 0.00 Mode :character
## Mean : 0.3502 Mean : 2.13
## 3rd Qu.: 0.0000 3rd Qu.: 1.00
## Max. :2404.0000 Max. :9572.00
```

```
## NA's :3
## user_created_at user_followers user_friends date
## Length:114213 Min. : 0 Min. : 0 Length:114213
## Class :character 1st Qu.: 106 1st Qu.: 179 Class :character
## Mode :character Median : 456 Median : 498 Mode :character
## Mean : 92012 Mean : 1429
## 3rd Qu.: 2007 3rd Qu.: 1188
## Max. :78061816 Max. :772902
## NA's :9 NA's :9
## language
## Length:114213
## Class :character
## Mode :character
##
##
##
##
```

Now I will check the dataset for missing or incorrect values that needs handling.

```
# retweet_count
tweets <- tweets %>%
  mutate(retweet_count = ifelse(retweet_count < 0, NA, retweet_count)) %>%
  mutate(retweet_count = replace_na(retweet_count, 0))

# favorite_count
tweets <- tweets %>%
  mutate(favorite_count = ifelse(favorite_count < 0, NA, favorite_count)) %>%
  mutate(favorite_count = replace_na(favorite_count, 0))

# user_followers
tweets <- tweets %>%
  mutate(user_followers = ifelse(user_followers < 0, NA, user_followers)) %>%
  mutate(user_followers = replace_na(user_followers, 0))

# user_friends
tweets <- tweets %>%
  mutate(user_friends = ifelse(user_friends < 0, NA, user_friends)) %>%
  mutate(user_friends = replace_na(user_friends, 0))

# user_created_at
tweets <- tweets %>%
  mutate(user_created_at = dmy_hm(user_created_at)) %>%
  filter(
    user_created_at >= as.POSIXct("2006-03-21"),
    user_created_at <= Sys.time()
  )

# date
tweets <- tweets %>%
  mutate(date = dmy_hm(date)) %>%
  filter(
    date >= as.POSIXct("2006-03-21"),
    date <= Sys.time()
  )
```

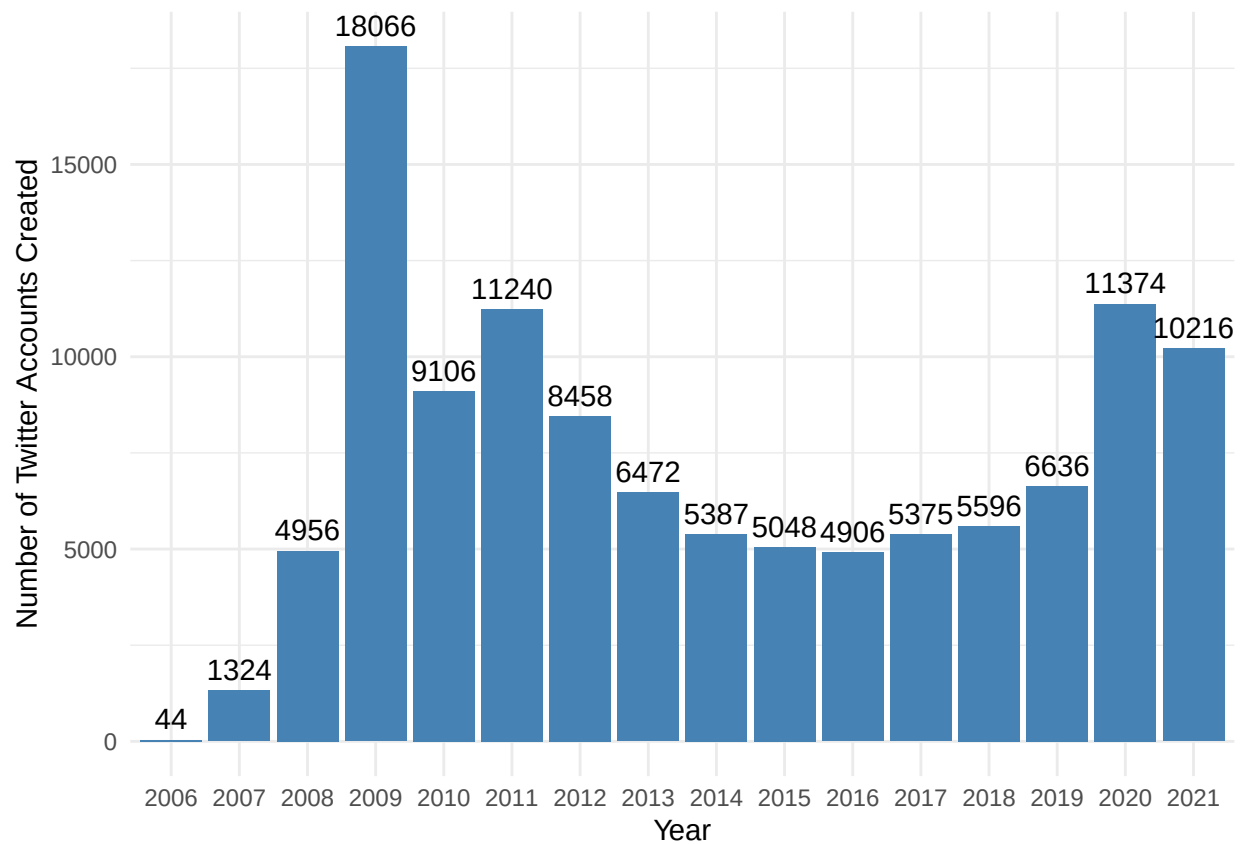
)

1.1 Visualisation

1.1.1 Number of Twitter Accounts Created per Year

```
# Create a new column for the year
tweets$year <- year(tweets$user_created_at)
tweets$year <- as.integer(tweets$year)
tweets$year <- as.factor(tweets$year)

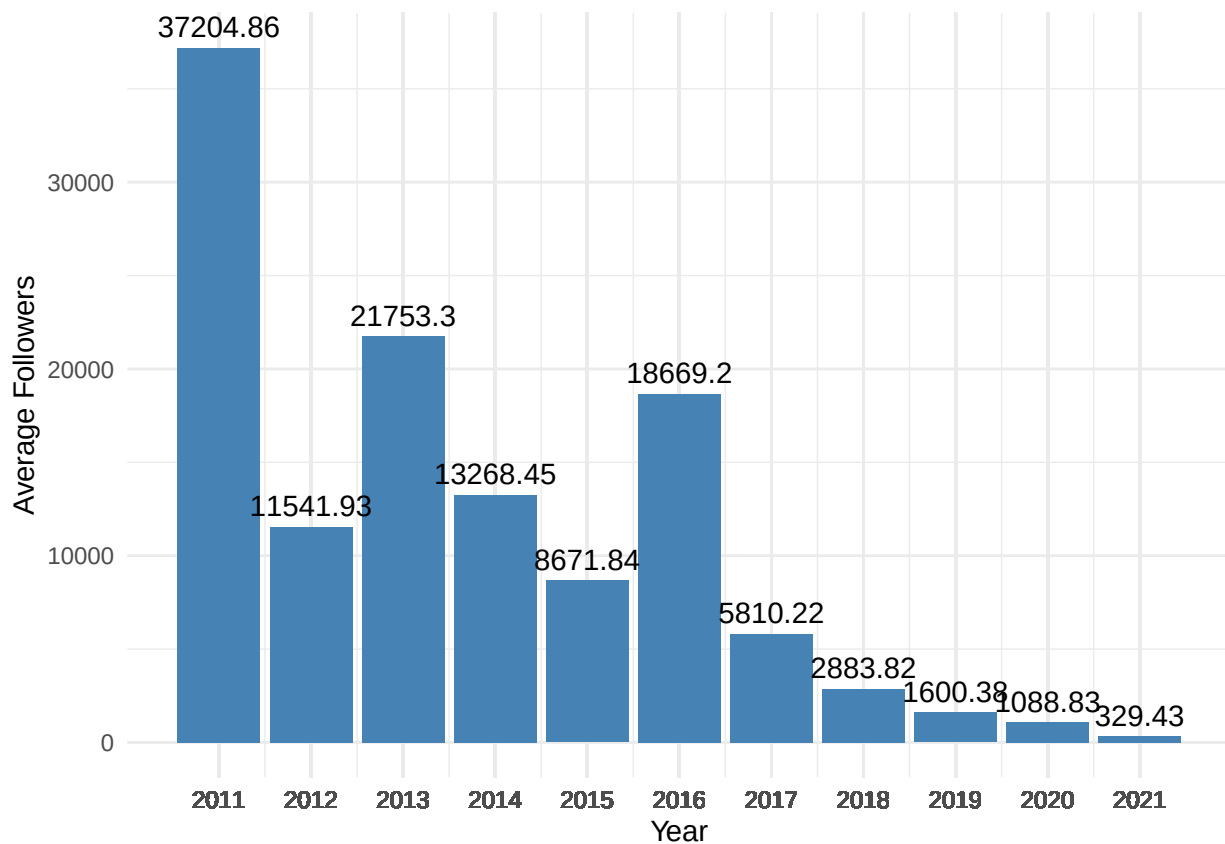
# Bar plot of the number of Twitter accounts per year
ggplot(tweets, aes(x = year)) +
  geom_bar(fill = "steelblue") +
  geom_text(
    stat = "count", aes(label = ..count..),
    position = position_dodge(width = 0.9),
    vjust = -0.5) +
  labs(x = "Year", y = "Number of Twitter Accounts Created") +
  theme_minimal()
```



1.1.2 Average Followers per Year

```
creators_after_2010 <- tweets
creators_after_2010 <- creators_after_2010 %>% mutate(year = as.numeric(as.character(year)))
creators_after_2010 <- creators_after_2010 %>% filter(year > 2010)

creators_after_2010 %>%
  filter(year > 2010) %>%
  group_by(year) %>%
  summarise(avg_followers = mean(user_followers, na.rm = TRUE)) %>%
  ggplot(aes(x = year, y = avg_followers)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(
    aes(label = round(avg_followers, 2)),
    vjust = -0.5) +
  scale_x_continuous(breaks = creators_after_2010$year) +
  labs(x = "Year", y = "Average Followers") +
  theme_minimal()
```



1.1.3 Top 10 Most Frequent Location of Users

```
top_locations <- tweets %>%
  select(user_screen_name, user_location) %>%
```



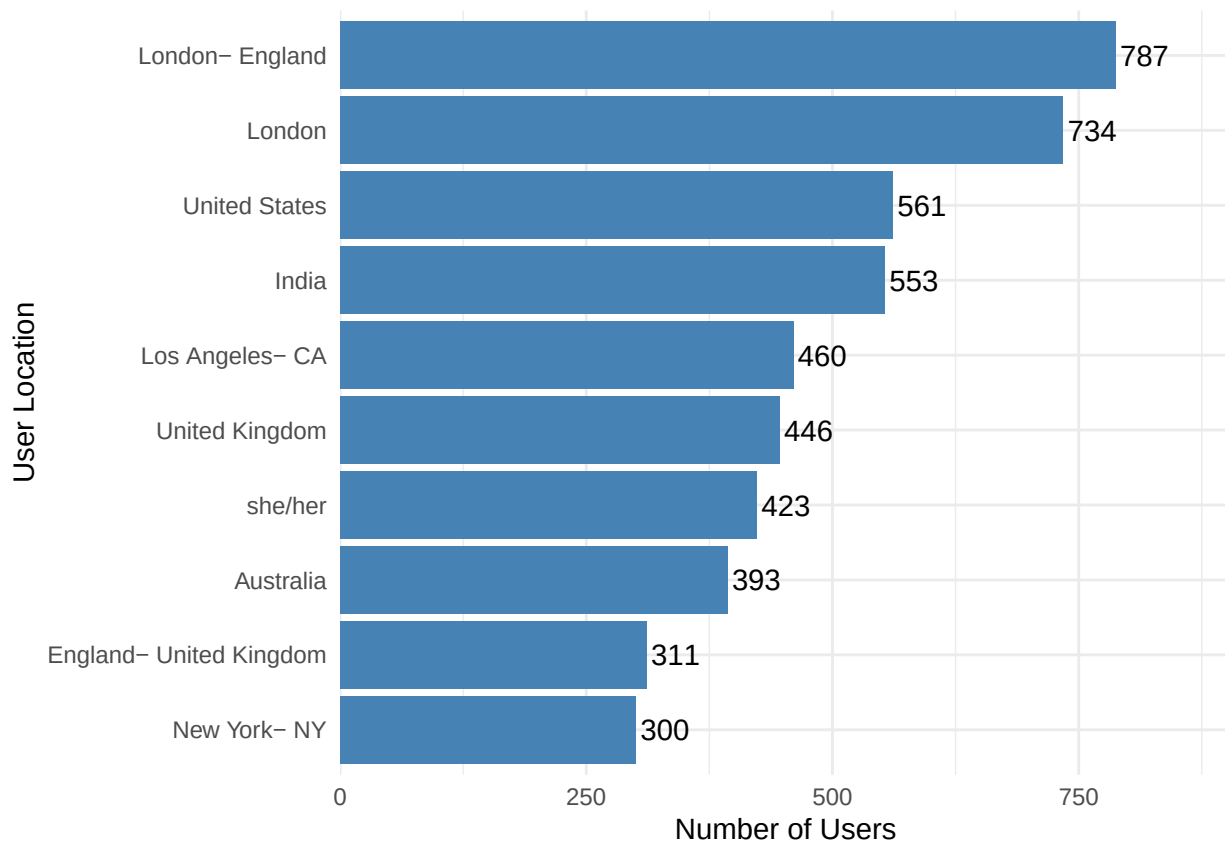
```

distinct() %>%
filter(!is.na(user_location), user_location != "") %>%
count(user_location, name="user_count") %>%
arrange(desc(user_count)) %>%
slice(1:10)

top_locations <- top_locations %>%
  mutate(user_location = fct_reorder(user_location, user_count, .desc = TRUE)) %>%
  mutate(user_location = fct_rev(user_location))

ggplot(top_locations, aes(x = user_location, y = user_count)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = user_count), hjust = -0.1, vjust = 0.5) +
  coord_flip() +
  labs(x = "User Location", y = "Number of Users") +
  theme_minimal() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))

```



1.1.4 Number of Tweets Associated with the Top 10 locations

```

tweets_in_top_10_locations <- tweets %>%
  filter(user_location %in% top_locations$user_location) %>%
  group_by(user_location) %>%

```

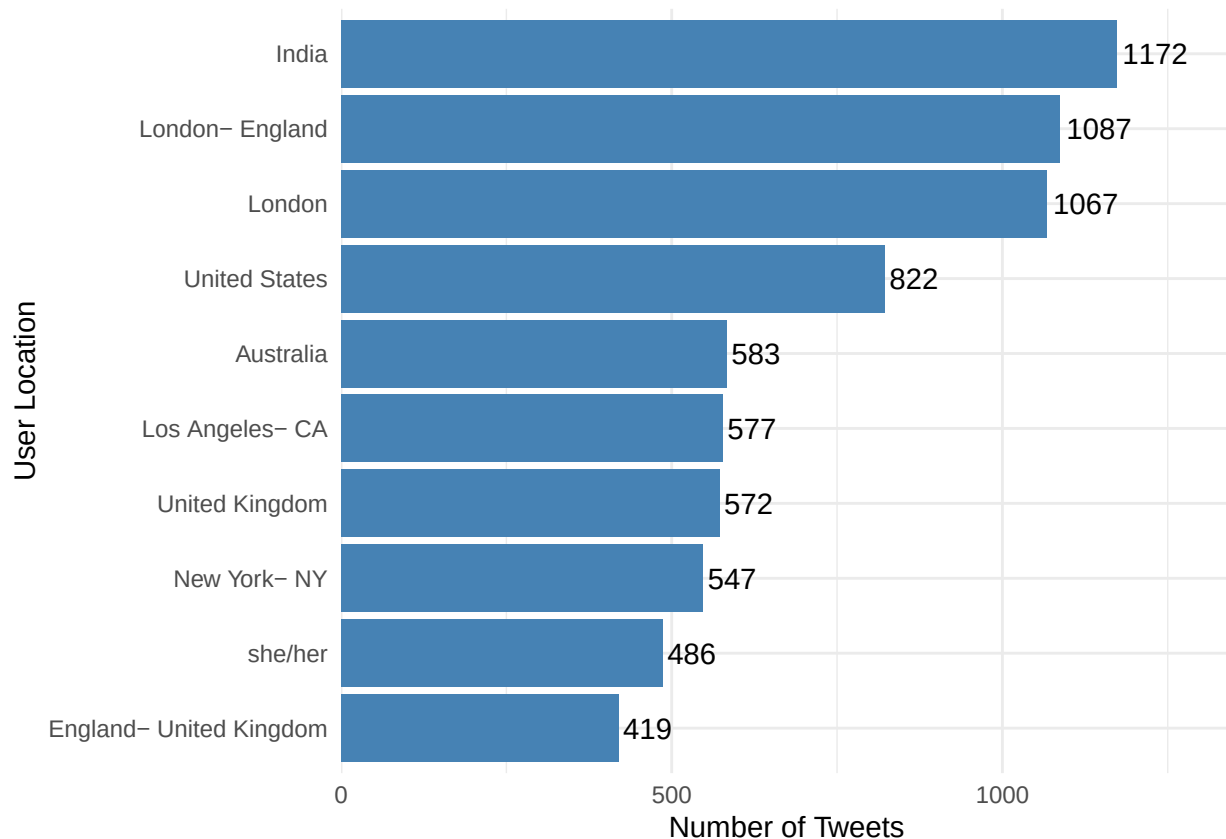
```

summarise(tweet_count = n()) %>%
  arrange(desc(tweet_count))

tweets_in_top_10_locations <- tweets_in_top_10_locations %>%
  mutate(user_location = fct_reorder(user_location, tweet_count, .desc = TRUE)) %>%
  mutate(user_location = fct_rev(user_location))

ggplot(tweets_in_top_10_locations, aes(x = user_location, y = tweet_count)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = tweet_count), hjust = -0.1, vjust = 0.5) +
  coord_flip() +
  labs(x = "User Location", y = "Number of Tweets") +
  theme_minimal() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))

```



From the last two visualisations, the odd values are “she/her”, and some locations are the geographically the same from a broader perspective, but some are more specific to cities. For example, “New York- NY”, “Los Angeles- CA”, and “United States”. The first two locations are cities in the United States, but there is also “United States” as a location.

1.1.5 Bar Chart of the Length of Tweets

```

tweets <- tweets %>%
  mutate(tweet_length = nchar(text)) %>%

```

```

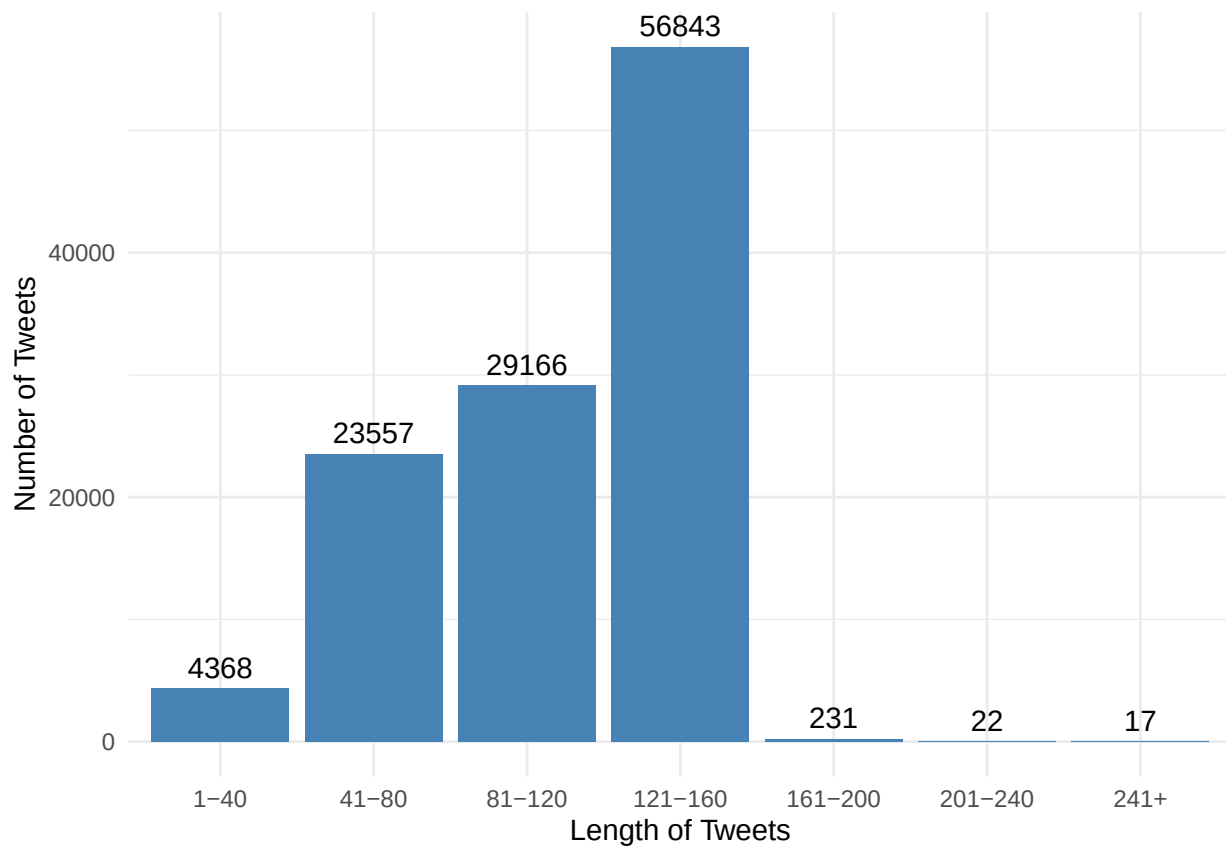
mutate(length_group = case_when(
  tweet_length >= 1 & tweet_length <= 40 ~ "1-40",
  tweet_length >= 41 & tweet_length <= 80 ~ "41-80",
  tweet_length >= 81 & tweet_length <= 120 ~ "81-120",
  tweet_length >= 121 & tweet_length <= 160 ~ "121-160",
  tweet_length >= 161 & tweet_length <= 200 ~ "161-200",
  tweet_length >= 201 & tweet_length <= 240 ~ "201-240",
  tweet_length >= 241 ~ "241+",
  TRUE ~ "Unknown"
))

length_group_counts <- tweets %>%
  count(length_group)

custom_order <- c("1-40", "41-80", "81-120", "121-160", "161-200", "201-240", "241+")
length_group_counts <- length_group_counts %>%
  mutate(length_group = factor(length_group, levels = custom_order))

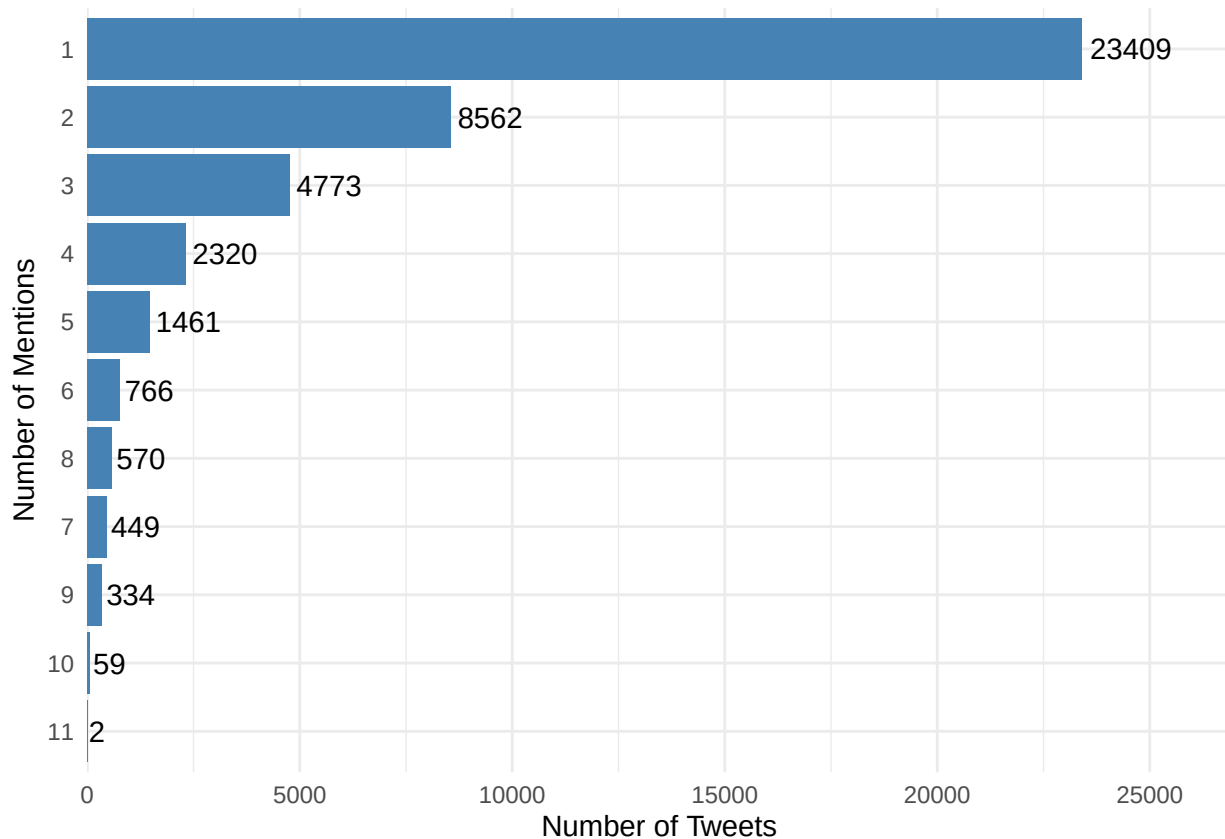
ggplot(length_group_counts, aes(x = length_group, y = n)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = n), vjust = -0.5) +
  labs(x = "Length of Tweets", y = "Number of Tweets") +
  theme_minimal()

```



1.1.6 Number of Tweets with Mentions and Its Count

```
tweets_with_mentions <- tweets %>%  
  filter(str_detect(text, "@[A-Za-z0-9_]+")) %>%  
  mutate(mentions_count = str_count(text, "@[A-Za-z0-9_]+"))  
  
tweets_with_mentions <- tweets_with_mentions %>%  
  group_by(mentions_count) %>%  
  summarise(tweet_count = n()) %>%  
  arrange(desc(tweet_count))  
  
tweets_with_mentions <- tweets_with_mentions %>%  
  mutate(mentions_count = as.factor(mentions_count)) %>%  
  mutate(mentions_count = fct_reorder(mentions_count, tweet_count, .desc = TRUE)) %>%  
  mutate(mentions_count = fct_rev(mentions_count))  
  
ggplot(tweets_with_mentions, aes(x = mentions_count, y = tweet_count)) +  
  geom_col(fill = "steelblue") +  
  coord_flip() +  
  geom_text(aes(label = tweet_count), hjust = -0.1, vjust = 0.5) +  
  labs(x = "Number of Mentions", y = "Number of Tweets") +  
  theme_minimal() +  
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))
```



```

number_of_tweets_with_mentions <- tweets %>%
  filter(str_detect(text, "@[A-Za-z0-9_]+")) %>%
  nrow()

number_of_tweets_with_3_or_more_mentions <- tweets %>%
  filter(str_detect(text, "@[A-Za-z0-9_]+")) %>%
  mutate(mentions_count = str_count(text, "@[A-Za-z0-9_]+")) %>%
  filter(mentions_count >= 3) %>%
  nrow()

cat("Number of tweets with mentions: ", number_of_tweets_with_mentions, "\n")

## Number of tweets with mentions: 42705

cat("Number of tweets with 3 or more mentions: ", number_of_tweets_with_3_or_more_mentions, "\n")

## Number of tweets with 3 or more mentions: 10734

```

1.1.7 Top 20 Most Frequent Term in Tweets

```

library(tidytext)

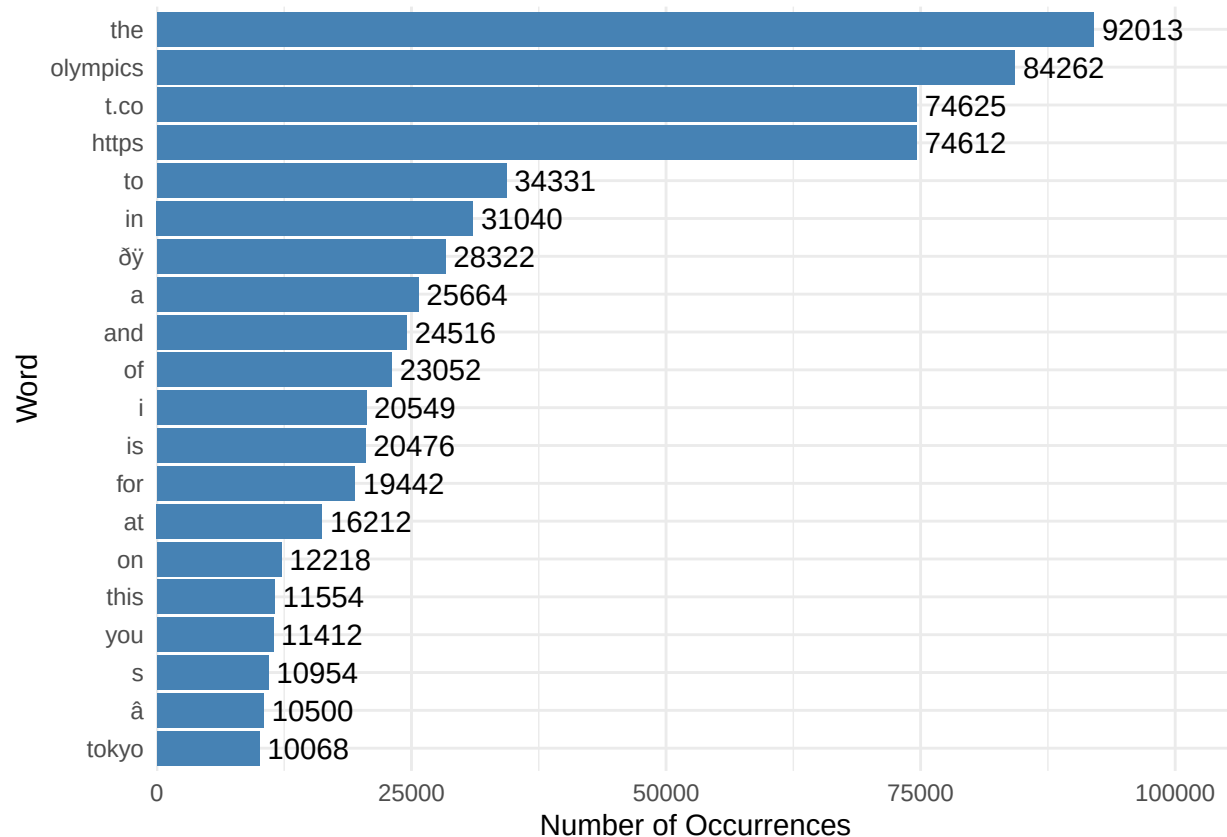
tweets_frequent_terms <- tweets %>%
  mutate(text = tolower(text)) %>%
  unnest_tokens(word, text) %>%
  count(word, sort = TRUE)

top_20_terms <- tweets_frequent_terms %>%
  slice(1:20) %>%
  mutate(word = fct_reorder(word, n)) %>%
  mutate(word = fct_rev(word))

max_count <- max(top_20_terms$n)

ggplot(top_20_terms, aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = n), hjust = -0.1, vjust = 0.5) +
  coord_flip() +
  labs(x = "Word", y = "Number of Occurrences") +
  theme_minimal() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))

```



1.1.8 Top 20 Most Frequent Non-Stopword Term in Tweets

```

tweets_frequent_terms_no_stopwords <- tweets %>%
  mutate(text = tolower(text)) %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words) %>%
  count(word, sort = TRUE)

top_20_terms_no_stopwords <- tweets_frequent_terms_no_stopwords %>%
  slice(1:20) %>%
  mutate(word = fct_reorder(word, n)) %>%
  mutate(word = fct_rev(word))

max_count_no_stopwords <- max(top_20_terms_no_stopwords$n)

ggplot(top_20_terms_no_stopwords, aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = n), hjust = -0.1, vjust = 0.5) +
  coord_flip() +
  labs(x = "Word", y = "Number of Occurrences") +
  theme_minimal() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))

```

